

Netflix AI Movie Recommendation System

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1 Project Overview

The Netflix AI Movie Recommendation System is a machine learning-powered web application that recommends movies based on content similarity. The system combines Natural Language Processing (NLP), machine learning techniques, and a modern web interface to provide users with personalized movie recommendations.

The application analyzes movie metadata such as movie titles, descriptions, and genres to identify similar movies and present recommendations through a Netflix-inspired user interface. Additionally, the system integrates with the TMDb (The Movie Database) API to fetch real movie posters and enhance the user experience.

2 Objectives

The primary objectives of this project are:

- Develop an intelligent movie recommendation system using machine learning techniques.
- Provide personalized movie suggestions based on movie content.
- Create an interactive and user-friendly Netflix-style web interface.
- Integrate external APIs to display real movie posters and metadata.
- Implement user authentication and watchlist functionality.

3 Dataset Description

The dataset contains information about:

- Movie Name
- Description
- Genre

Movie data was collected using the TMDb API and processed before model training.

4 Data Preprocessing

The following preprocessing techniques were applied:

- Lowercasing
- HTML Tag Removal
- URL Removal
- Punctuation Removal
- Stop Word Removal
- Tokenization
- Spelling Correction
- Stemming
- Lemmatization

5 Machine Learning Methodology

5.1 Feature Extraction

Movie descriptions were converted into numerical vectors using NLP techniques such as:

- TF-IDF Vectorization
- Sentence Embeddings

5.2 Similarity Calculation

Cosine Similarity was used to measure similarity between movie vectors.

$$\text{Cosine Similarity} = \frac{A \cdot B}{\|A\| \|B\|}$$

where:

- A = Movie Vector A
- B = Movie Vector B

6 Technology Stack

Component	Technology
Programming Language	Python
Frontend Framework	Streamlit
Machine Learning	Scikit-Learn
Data Processing	Pandas, NumPy
NLP Libraries	NLTK, spaCy
Recommendation Algorithm	Cosine Similarity
API Integration	TMDB API
Authentication	bcrypt
Deployment	Streamlit Cloud

Table 1: Technology Stack

7 System Workflow

1. Collect movie data using the TMDB API.
2. Perform data cleaning and preprocessing.
3. Generate TF-IDF vectors or sentence embeddings.
4. Calculate cosine similarity.
5. Store processed data and embeddings.
6. User selects a movie.
7. Recommendation engine finds similar movies.
8. TMDB API fetches movie posters.
9. Results are displayed through the Streamlit interface.

8 Features

- AI-Powered Movie Recommendations
- Netflix-Style User Interface
- Real-Time Movie Posters
- User Authentication
- Personalized Watchlist
- Trending Movies Section
- Streamlit Cloud Deployment

9 Future Enhancements

- Trailer integration using YouTube API
- User rating and review system
- Personalized recommendations based on user history
- Deep learning recommendation models
- Mobile application development
- Firebase or MongoDB integration

10 Conclusion

The Netflix AI Movie Recommendation System demonstrates the practical application of Machine Learning, Natural Language Processing, and Web Development technologies. By combining content-based recommendation techniques with a modern web interface and TMDB API integration, the system delivers an engaging and intelligent movie recommendation platform suitable for academic, portfolio, and real-world applications.